

MIPT & YSDA · AUTUMN 2026

Deep Generative Models

Lecture 2

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Recap of Previous Lecture

We're given a **finite** number of i.i.d. samples $\{\mathbf{x}_i\}_{i=1}^n \subset \mathbb{R}^m$ drawn from an **unknown** distribution $p_{\text{data}}(\mathbf{x})$.

Objective

Our aim is to estimate a distribution $p_{\text{data}}(\mathbf{x})$ that allows us to:

- **Sampling (generation):** sample $\mathbf{x} \sim p_{\text{data}}(\mathbf{x})$.
- **Density evaluation:** evaluate $p_{\text{data}}(\mathbf{x})$ on novel data ("How likely is an object $\mathbf{x}$?").

What is a Divergence?

- $D(\pi \| p) \geq 0$ for all $\pi, p \in \mathcal{P}$
- $D(\pi \| p) = 0$ if and only if $\pi \equiv p$

Divergence Minimization Problem

$$\min_{\theta} D(p_{\text{data}} \| p_{\theta})$$

Recap of Previous Lecture

Forward KL

$$D_{\text{KL}}(p_{\text{data}} \| p_{\theta}) = \int p_{\text{data}}(\mathbf{x}) \log \frac{p_{\text{data}}(\mathbf{x})}{p_{\theta}(\mathbf{x})} d\mathbf{x} \rightarrow \min_{\theta}$$

Reverse KL

$$D_{\text{KL}}(p_{\theta} \| p_{\text{data}}) = \int p_{\theta}(\mathbf{x}) \log \frac{p_{\theta}(\mathbf{x})}{p_{\text{data}}(\mathbf{x})} d\mathbf{x} \rightarrow \min_{\theta}$$

Maximum Likelihood Estimation (MLE)

Let $\{\mathbf{x}_i\}_{i=1}^n$ be i.i.d. observed samples.

$$\theta^* = \arg \max_{\theta} \prod_{i=1}^n p_{\theta}(\mathbf{x}_i) = \arg \max_{\theta} \sum_{i=1}^n \log p_{\theta}(\mathbf{x}_i).$$

Maximum likelihood estimation is equivalent to minimizing a Monte Carlo estimate of the forward KL divergence.

Recap of Previous Lecture

Likelihood as a Product of Conditionals

For $\mathbf{x} = (x_1, \dots, x_m)$, $\mathbf{x}_{1:j} = (x_1, \dots, x_j)$,

$$p_{\theta}(\mathbf{x}) = \prod_{j=1}^m p_{\theta}(x_j | \mathbf{x}_{1:j-1}); \quad \log p_{\theta}(\mathbf{x}) = \sum_{j=1}^m \log p_{\theta}(x_j | \mathbf{x}_{1:j-1})$$

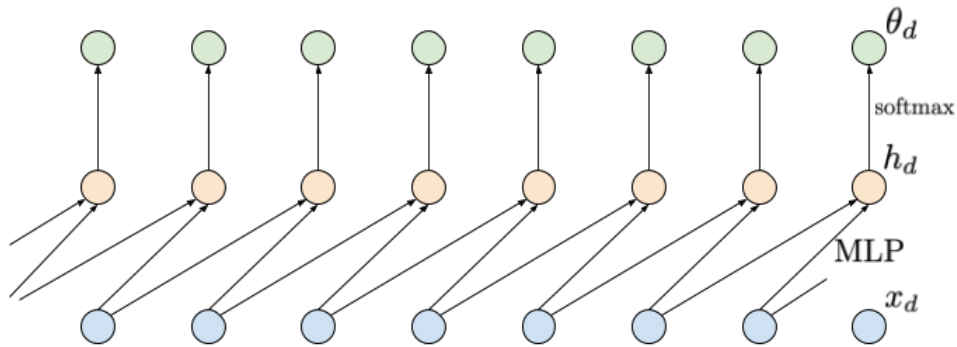
$$\theta^* = \arg \max_{\theta} \sum_{i=1}^n \left[\sum_{j=1}^m \log p_{\theta}(x_{ij} | \mathbf{x}_{i,1:j-1}) \right]$$

Sampling (Ancestral)

1. Sample $\hat{x}_1 \sim p_{\theta}(x_1)$, $\hat{x}_2 \sim p_{\theta}(x_2 | \hat{x}_1)$, $\dots$, $\hat{x}_m \sim p_{\theta}(x_m | \hat{\mathbf{x}}_{1:m-1})$.
2. Return $\hat{\mathbf{x}} = (\hat{x}_1, \hat{x}_2, \dots, \hat{x}_m)$.

Recap of Previous Lecture

Autoregressive Models: MLP



Autoregressive Models: ImageGPT

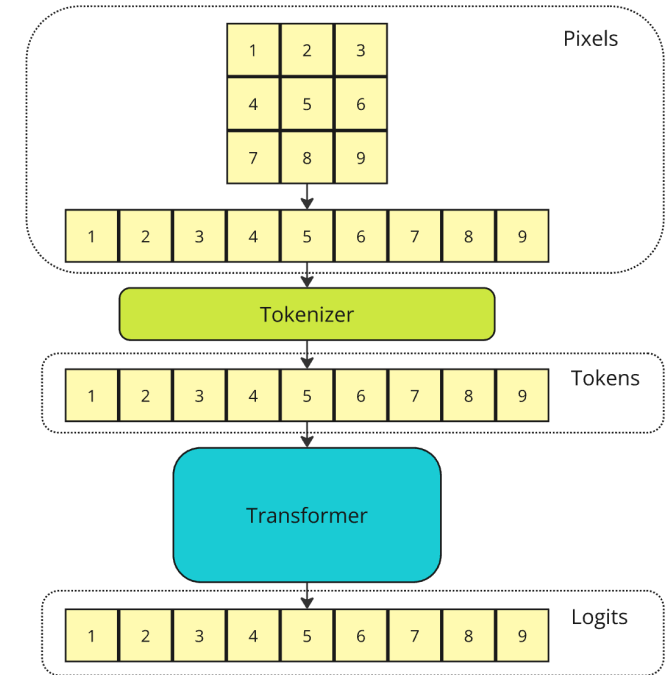
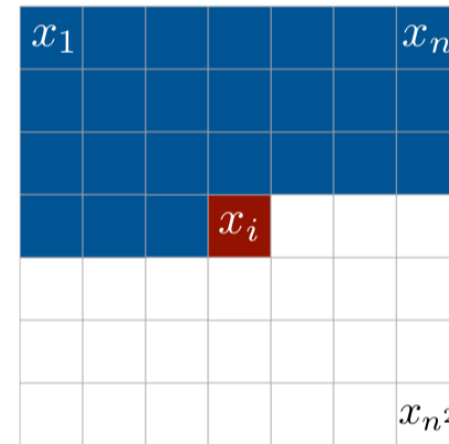


Image credit: https://jmtomczak.github.io/blog/2/2_ARM.html

Oord A., Kalchbrenner N., Kavukcuoglu K. Pixel Recurrent Neural Networks, 2016

Chen M. et al. Generative Pretraining from Pixels, 2020

Outline

01 Normalizing Flows (NF)

02 NF Examples

- Linear NF

- Gaussian Autoregressive NF

- Coupling Layer (RealNVP)

03 Latent Variable Models (LVM)

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01 Normalizing Flows (NF)

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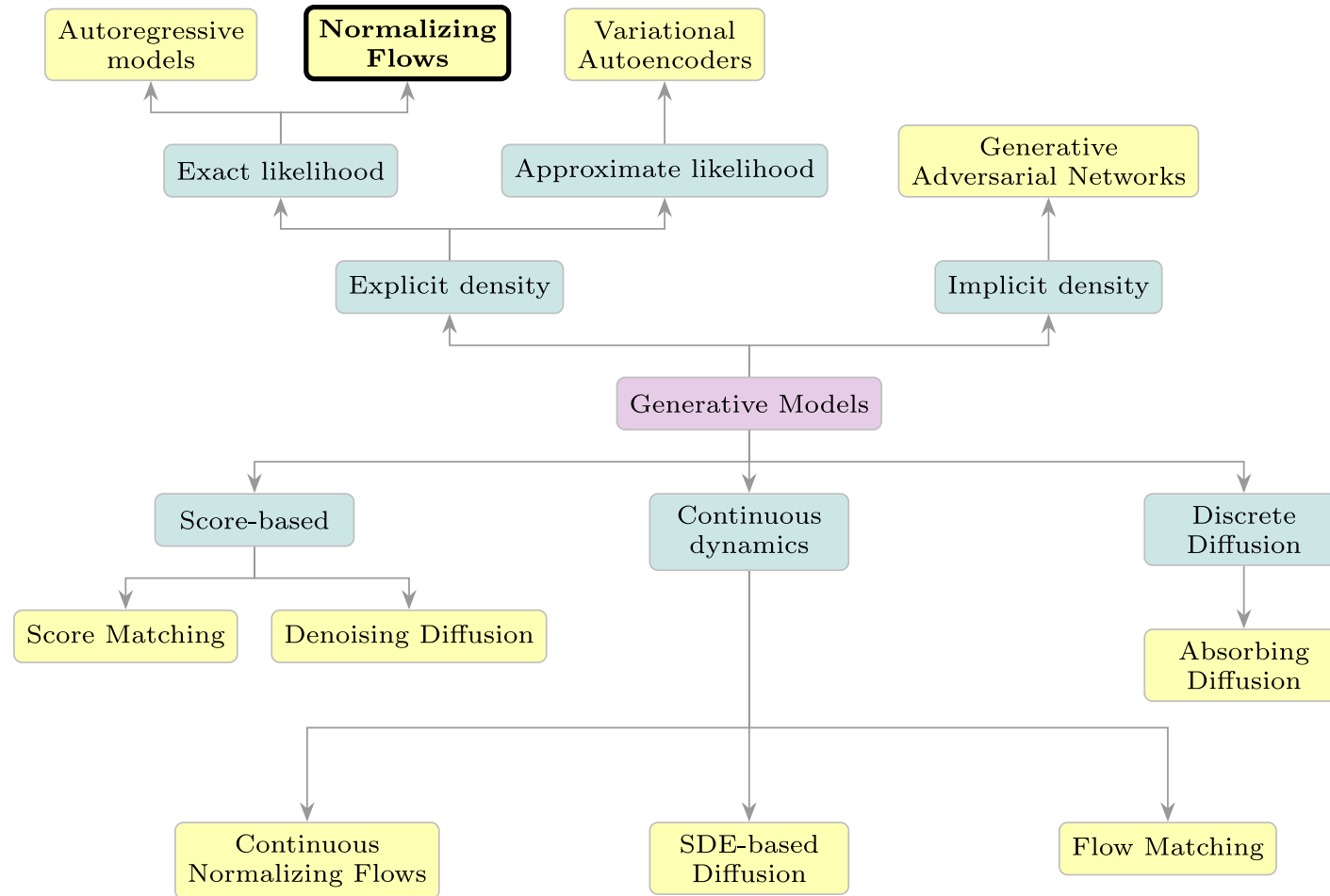
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Generative Models Taxonomy



Normalizing Flows: Prerequisites

Jacobian Matrix

Let $\mathbf{f} : \mathbb{R}^m \rightarrow \mathbb{R}^m$ be a differentiable function.

$$\mathbf{z} = \mathbf{f}(\mathbf{x})$$

$$\mathbf{J} = \frac{\partial \mathbf{z}}{\partial \mathbf{x}} = \begin{pmatrix} \frac{\partial z_1}{\partial x_1} & \cdots & \frac{\partial z_1}{\partial x_m} \\ \vdots & \ddots & \vdots \\ \frac{\partial z_m}{\partial x_1} & \cdots & \frac{\partial z_m}{\partial x_m} \end{pmatrix} \in \mathbb{R}^{m \times m}$$

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Change of Variables Theorem (CoV)

Let $\mathbf{x} \in \mathbb{R}^m$ be a random vector with density $p(\mathbf{x})$, and let $\mathbf{f} : \mathbb{R}^m \rightarrow \mathbb{R}^m$ be a C^1 -diffeomorphism ($\mathbf{f}$ and $\mathbf{f}^{-1}$ are continuously differentiable mappings). If $\mathbf{z} = \mathbf{f}(\mathbf{x})$, then

$$p(\mathbf{x}) = p(\mathbf{z}) |\det(\mathbf{J}_{\mathbf{f}})| = p(\mathbf{z}) \left| \det \left(\frac{\partial \mathbf{z}}{\partial \mathbf{x}} \right) \right|$$

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Jacobian Determinant

Inverse Function Theorem

If the function $\mathbf{f}$ is invertible and its Jacobian is continuous and non-singular, then

$$\mathbf{J}_{\mathbf{f}^{-1}} = \mathbf{J}_{\mathbf{f}}^{-1}.$$

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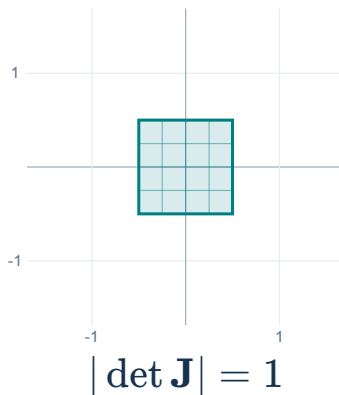
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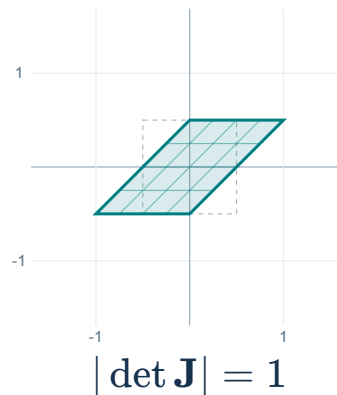
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The determinant of the Jacobian $\mathbf{J} = \frac{\partial \mathbf{f}_{\theta}(\mathbf{x})}{\partial \mathbf{x}}$ quantifies how the volume is changed by the transformation.

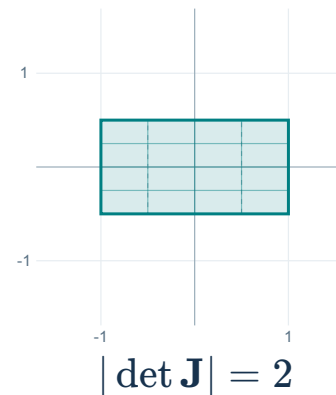
Identity



Shear



Stretch



$$\mathbf{z} = \mathbf{J}\mathbf{x}, \quad \mathbf{J} = \begin{pmatrix} a & b \\ 0 & 1 \end{pmatrix}$$

Shear preserves area; stretching by 2 doubles it.

$$p(\mathbf{z}) = p(\mathbf{x}) / |\det \mathbf{J}| \text{ — probability mass is preserved.}$$

Fitting Normalizing Flows

MLE Problem

$$p_{\theta}(\mathbf{x}) = p(\mathbf{z}) \left| \det \left(\frac{\partial \mathbf{z}}{\partial \mathbf{x}} \right) \right| = p(\mathbf{f}_{\theta}(\mathbf{x})) \left| \det \left(\frac{\partial \mathbf{f}_{\theta}(\mathbf{x})}{\partial \mathbf{x}} \right) \right|$$

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Inference

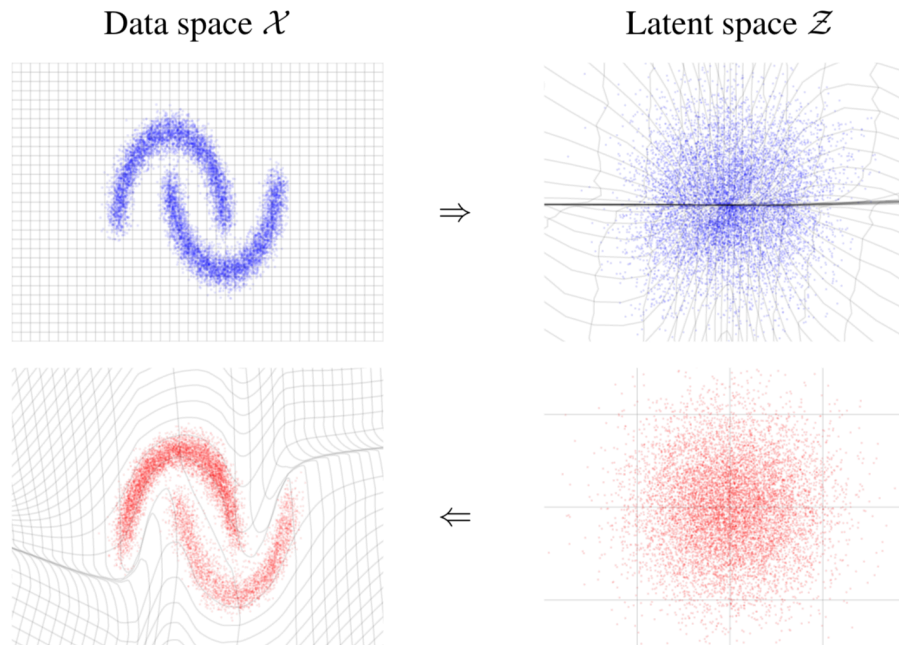
$$x \sim \hat{p}_X$$

$$z = f(x)$$

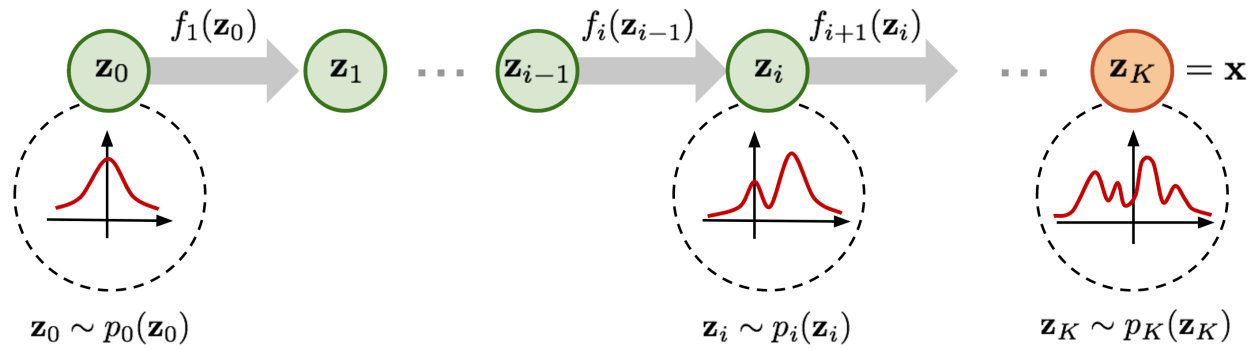
Generation

$$z \sim p_Z$$

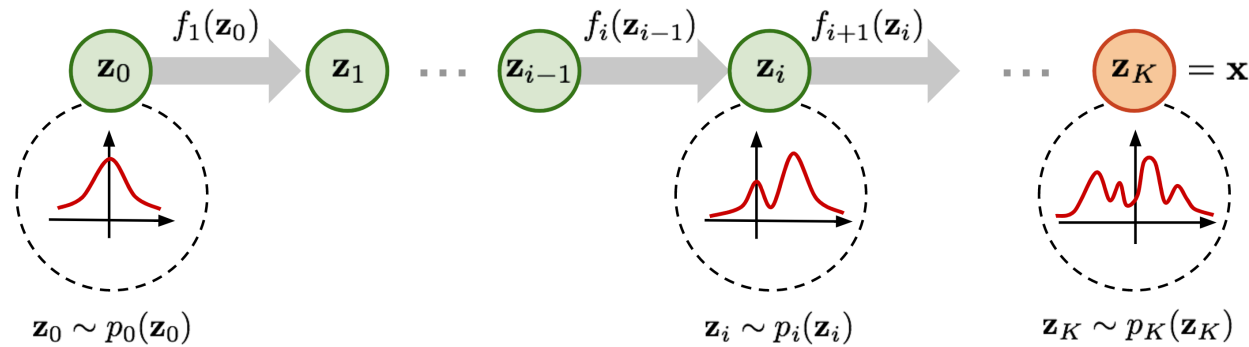
$$x = f^{-1}(z)$$



Composition of Normalizing Flows



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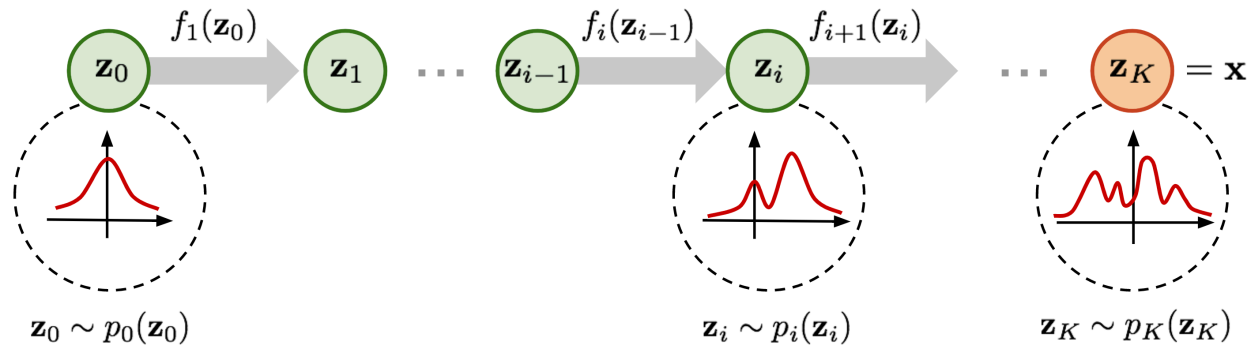


Theorem

If every $\{\mathbf{f}_k\}_{k=1}^K$ satisfies the conditions of the change-of-variables theorem, then the composition $\mathbf{f}(\mathbf{x}) = \mathbf{f}_K \circ \dots \circ \mathbf{f}_1(\mathbf{x})$ also satisfies them.

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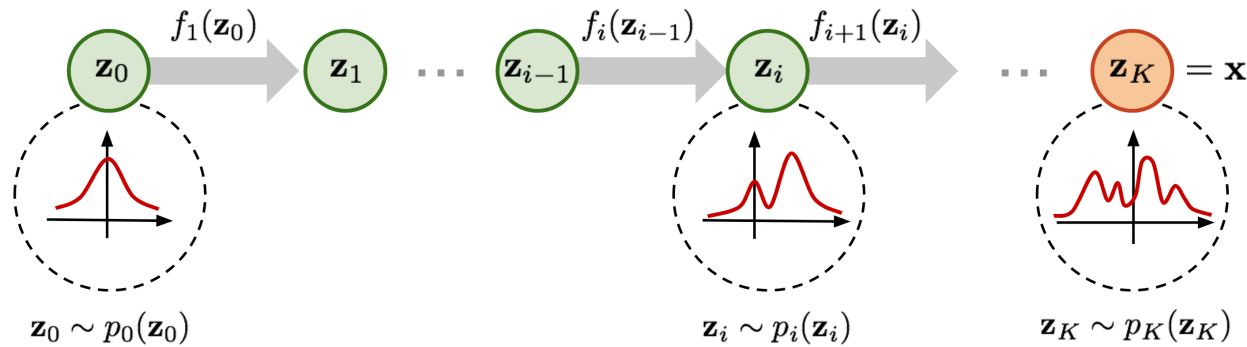


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$$\begin{aligned}
 p_{\theta}(\mathbf{x}) &= p(\mathbf{f}(\mathbf{x})) \left| \det \left(\frac{\partial \mathbf{f}(\mathbf{x})}{\partial \mathbf{x}} \right) \right| = p(\mathbf{f}(\mathbf{x})) \left| \det \left(\frac{\partial \mathbf{f}_K}{\partial \mathbf{f}_{K-1}} \cdots \frac{\partial \mathbf{f}_1}{\partial \mathbf{x}} \right) \right| \\
 &= p(\mathbf{f}(\mathbf{x})) \prod_{k=1}^K \left| \det \left(\frac{\partial \mathbf{f}_k}{\partial \mathbf{f}_{k-1}} \right) \right| = p(\mathbf{f}(\mathbf{x})) \prod_{k=1}^K |\det(\mathbf{J}_{\mathbf{f}_k})|
 \end{aligned}$$

Normalizing Flows (NF)

$$\log p_{\theta}(\mathbf{x}) = \log p(\mathbf{f}_{\theta}(\mathbf{x})) + \log |\det(\mathbf{J}_{\mathbf{f}})|$$

Definition

A normalizing flow is a C^1 -diffeomorphism that transforms data $\mathbf{x}$ to noise $\mathbf{z}$.

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- **Normalizing** refers to mapping samples from $p_{\text{data}}(\mathbf{x})$ to a base distribution $p(\mathbf{z})$.
- **Flow** describes the sequence of transformations from base noise to data.

$$\mathbf{z} = \mathbf{f}_K \circ \dots \circ \mathbf{f}_1(\mathbf{x}); \quad \mathbf{x} = \mathbf{f}_1^{-1} \circ \dots \circ \mathbf{f}_K^{-1}(\mathbf{z})$$

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We focus on **continuous** variables; **discrete flows** have not become mainstream.

Normalizing Flows

Use a fixed base distribution $p(\mathbf{z})$, e.g. $\mathcal{N}(0, \mathbf{I})$.

Training

1. Sample a minibatch $\mathbf{x}_i \sim p_{\text{data}}(\mathbf{x})$, $i = 1, \dots, n$.
2. Compute $\mathbf{z}_i = \mathbf{f}_{\theta}(\mathbf{x}_i)$ and $\log |\det \mathbf{J}_{\mathbf{f}}(\mathbf{x}_i)|$.
3. Compute loss

$$\mathcal{L} = -\frac{1}{n} \sum_{i=1}^n [\log p(\mathbf{z}_i) + \log |\det \mathbf{J}_{\mathbf{f}}(\mathbf{x}_i)|] .$$

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Sampling

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Requirements

- **Training:** efficient computation of $\log |\det \mathbf{J}_{\mathbf{f}}|$, where $\mathbf{J}_{\mathbf{f}} = \frac{\partial \mathbf{f}_{\theta}(\mathbf{x})}{\partial \mathbf{x}}$.
- **Sampling:** efficient inversion of the transformation $\mathbf{f}_{\theta}$.

What's the computational complexity of evaluating this determinant?

Sampling

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Jacobian Structure

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The principal computational challenge is evaluating the Jacobian determinant.

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What is $\det(\mathbf{J})$ in These Cases?

Consider a linear layer $\mathbf{z} = \mathbf{W}\mathbf{x}$, $\mathbf{W} \in \mathbb{R}^{m \times m}$.

1. $\mathbf{z}$ is a permutation of $\mathbf{x}$.

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3. z_j depends only on $\mathbf{x}_{1:j}$ (autoregressive dependency).

Linear Normalizing Flows

$$\mathbf{z} = \mathbf{f}_\theta(\mathbf{x}) = \mathbf{W}\mathbf{x}, \quad \mathbf{W} \in \mathbb{R}^{m \times m}, \quad \mathbf{J}_f = \mathbf{W}$$

Computing $\mathbf{f}_\theta^{-1}(\mathbf{z})$ means solving $\mathbf{W}\mathbf{x} = \mathbf{z}$: $O(m^3)$ for a general dense matrix.

Diagonal systems cost $O(m)$; triangular systems cost $O(m^2)$.

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Continuous Parameterization

There is no continuous surjective map from $\mathbb{R}^{m^2}$ onto the set of all invertible $m \times m$ matrices.

Explanation: a continuous path from $\det \mathbf{W} > 0$ to $\det \mathbf{W} < 0$ must cross $\det \mathbf{W} = 0$, where invertibility is lost. A continuous parameterization that guarantees invertibility must therefore keep the determinant sign fixed.

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Structured Factors

Structured factorizations (e.g. LU or QR) simplify determinant and inverse calculations. Keep triangular diagonals nonzero to preserve invertibility.

Learn the factors directly during training.

Outline

01 Normalizing Flows (NF)

02 NF Examples

Linear NF

Gaussian Autoregressive NF

Coupling Layer (RealNVP)

03 Latent Variable Models (LVM)

Gaussian Autoregressive Model

Consider the autoregressive model:

$$p_{\boldsymbol{\theta}}(\mathbf{x}) = \prod_{j=1}^m p_{\boldsymbol{\theta}}(x_j | \mathbf{x}_{1:j-1}), \quad p_{\boldsymbol{\theta}}(x_j | \mathbf{x}_{1:j-1}) = \mathcal{N}(\mu_{j,\boldsymbol{\theta}}(\mathbf{x}_{1:j-1}), \sigma_{j,\boldsymbol{\theta}}^2(\mathbf{x}_{1:j-1}))$$

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- The Jacobian matrix of this transformation is triangular.

Gaussian Autoregressive NF: Two Directions

Sampling: sequential

$$x_j = \mu_{j,\theta}(\mathbf{x}_{1:j-1}) + \sigma_{j,\theta}(\mathbf{x}_{1:j-1})z_j$$

$$\mu_{j,\theta}(\mathbf{x}_{1:j-1}) = \sum_{i<j} x_i, \quad \sigma_{j,\theta} = 1 \quad \text{A fixed toy Gaussian AR flow}$$

Density evaluation: parallel

$$z_j = \frac{x_j - \mu_{j,\theta}(\mathbf{x}_{1:j-1})}{\sigma_{j,\theta}(\mathbf{x}_{1:j-1})}$$

Sampling: sequential

$$x_j = z_j + \sum_{i<j} x_i$$

x_1	x_2	x_3	x_4
-1.000	-0.500	-0.500	-2.500
step 1	step 2	step 3	step 4

Each coordinate needs the previous outputs.

Density evaluation: parallel

$$z_j = x_j - \sum_{i<j} x_i$$

z_1	z_2	z_3	z_4
-1.000	0.500	1.000	-0.500
same step	same step	same step	same step

All prefixes are already available in the observed vector.

The dependency structure is unchanged. What differs is which values are already known.

$\log p_{\theta}(\mathbf{x}) = \log p(\mathbf{z}) + \log |\det \mathbf{J}_{\mathbf{f}}|$; here the Jacobian has unit diagonal, so the log-determinant is zero.

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RealNVP

Split $\mathbf{x}$ and $\mathbf{z}$ into two parts:

$$\mathbf{x} = [\mathbf{x}_1, \mathbf{x}_2] = [\mathbf{x}_{1:d}, \mathbf{x}_{d+1:m}]; \quad \mathbf{z} = [\mathbf{z}_1, \mathbf{z}_2] = [\mathbf{z}_{1:d}, \mathbf{z}_{d+1:m}]$$

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$$\begin{cases} \mathbf{x}_1 = \mathbf{z}_1 \\ \mathbf{x}_2 = \mathbf{z}_2 \odot \boldsymbol{\sigma}_\theta(\mathbf{z}_1) + \boldsymbol{\mu}_\theta(\mathbf{z}_1) \end{cases}$$

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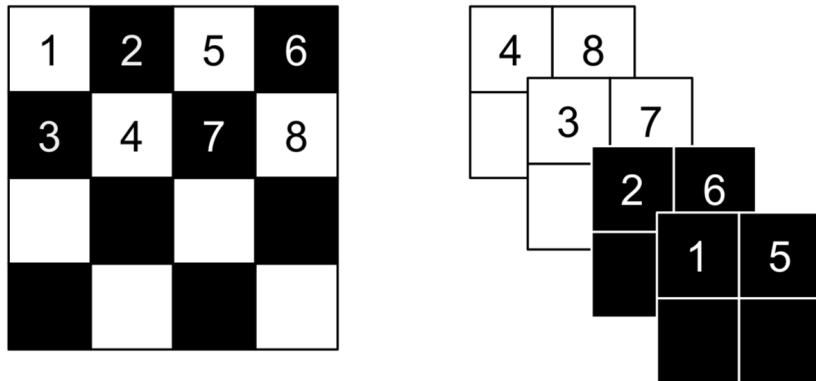
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Image Partitioning



- Checkerboard ordering corresponds to masking.
- Channelwise ordering relies on splitting.

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$$\det \left(\frac{\partial \mathbf{z}}{\partial \mathbf{x}} \right) = \det \begin{pmatrix} \mathbf{I}_d & 0_{d \times m-d} \\ \frac{\partial \mathbf{z}_2}{\partial \mathbf{x}_1} & \frac{\partial \mathbf{z}_2}{\partial \mathbf{x}_2} \end{pmatrix}$$

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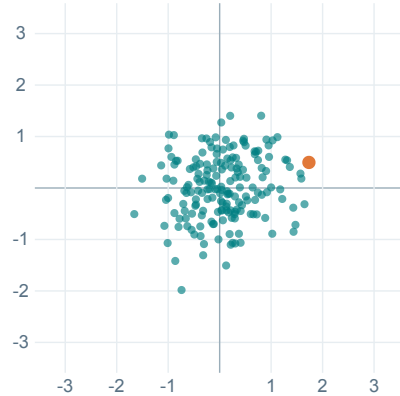
Gaussian AR NF

$$\begin{aligned} \mathbf{x} = \mathbf{f}_{\theta}^{-1}(\mathbf{z}) &\Rightarrow \mathbf{x}_j = \sigma_{j,\theta}(\mathbf{x}_{1:j-1}) \cdot \mathbf{z}_j + \mu_{j,\theta}(\mathbf{x}_{1:j-1}) \\ \mathbf{z} = \mathbf{f}_{\theta}(\mathbf{x}) &\Rightarrow \mathbf{z}_j = (\mathbf{x}_j - \mu_{j,\theta}(\mathbf{x}_{1:j-1})) \cdot \frac{1}{\sigma_{j,\theta}(\mathbf{x}_{1:j-1})}. \end{aligned}$$

How can the RealNVP layer be derived as a special instance of the Gaussian autoregressive NF?

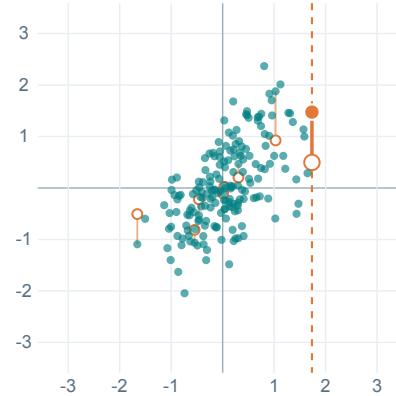
RealNVP: Explore a Coupling Layer

Base noise



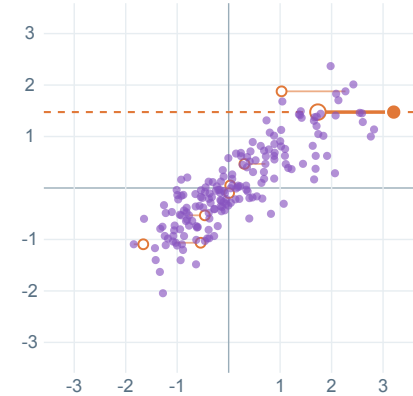
$$\mathbf{z} \sim \mathcal{N}(0, 0.65^2 \mathbf{I})$$

Layer 1: vertical motion



$$y_1 = z_1, \quad y_2 = \sigma(z_1)z_2 + \mu(z_1)$$

Layer 2: horizontal motion



$$x_2 = y_2, \quad x_1 = \sigma(y_2)y_1 + \mu(y_2)$$

$$\sigma(t) = e^{0.4\lambda \tanh(t)}, \quad \mu(t) = 0.8\lambda \tanh(t), \quad \lambda = 1.0 \quad \text{Dashed: fixed coordinate. Hollow: previous position.}$$

$$\frac{\partial \mathbf{y}}{\partial \mathbf{z}} = \begin{pmatrix} 1 & 0 \\ * & \sigma(z_1) \end{pmatrix}$$

$$\frac{\partial \mathbf{x}}{\partial \mathbf{y}} = \begin{pmatrix} \sigma(y_2) & * \\ 0 & 1 \end{pmatrix}$$

Triangular Jacobians.
Positive, explicit scales.

The conditioner stays fixed: undo the shift, divide by the scale. For density evaluation, negate the generative log-determinant.

TarFlow: Autoregression over Patches

Replace scalar coordinates by N image patches $\mathbf{x}_j \in \mathbb{R}^d$. Keep the same dependence on the **previous patches** $\mathbf{x}_{1:j-1}$.

Density Evaluation

$$\mathbf{z}_j = (\mathbf{x}_j - \boldsymbol{\mu}_{j,\theta}(\mathbf{x}_{1:j-1})) \odot \frac{1}{\boldsymbol{\sigma}_{j,\theta}(\mathbf{x}_{1:j-1})}.$$

All input patches are observed: compute every $\mathbf{z}_j$ **in parallel** within a block.

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Earlier output patches must be generated first: compute $\mathbf{x}_j$ **sequentially**.

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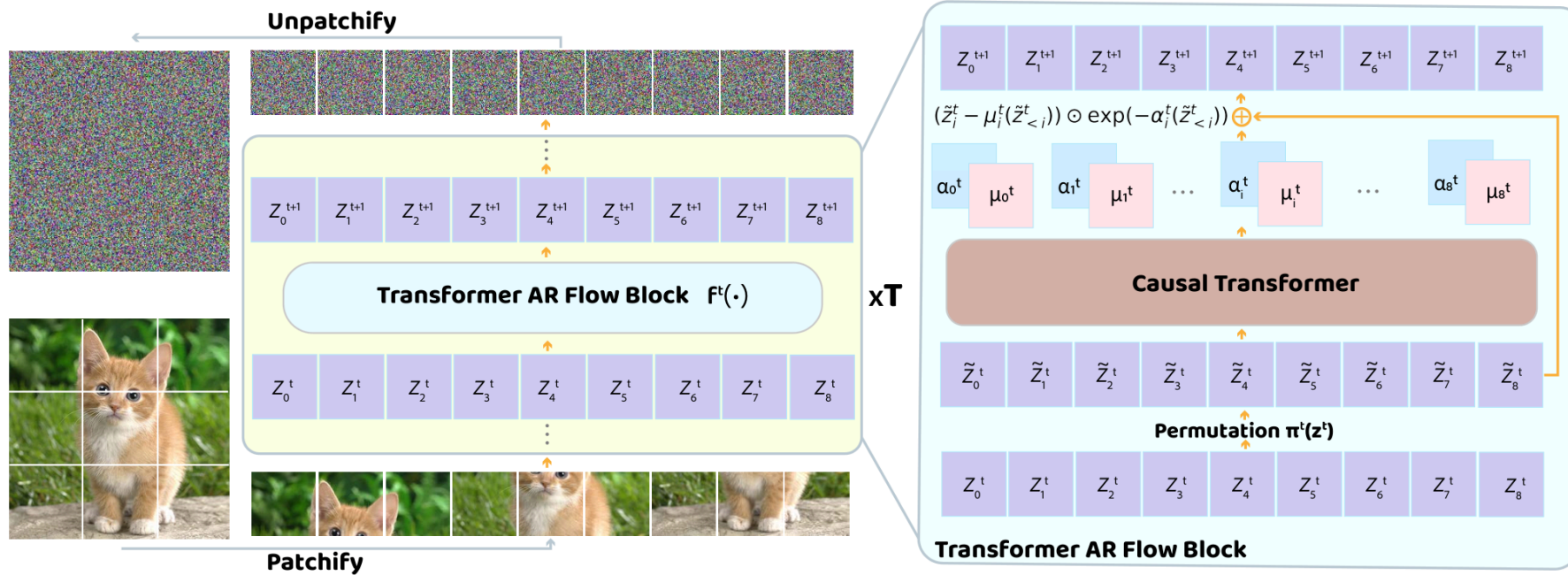
Earlier output patches must be generated first: compute $\mathbf{x}_j$ **sequentially**.

The Jacobian stays **triangular**; positive componentwise scales ensure invertibility.

Here $j = 2, \dots, N$; the first patch is unchanged: $\mathbf{z}_1 = \mathbf{x}_1$.

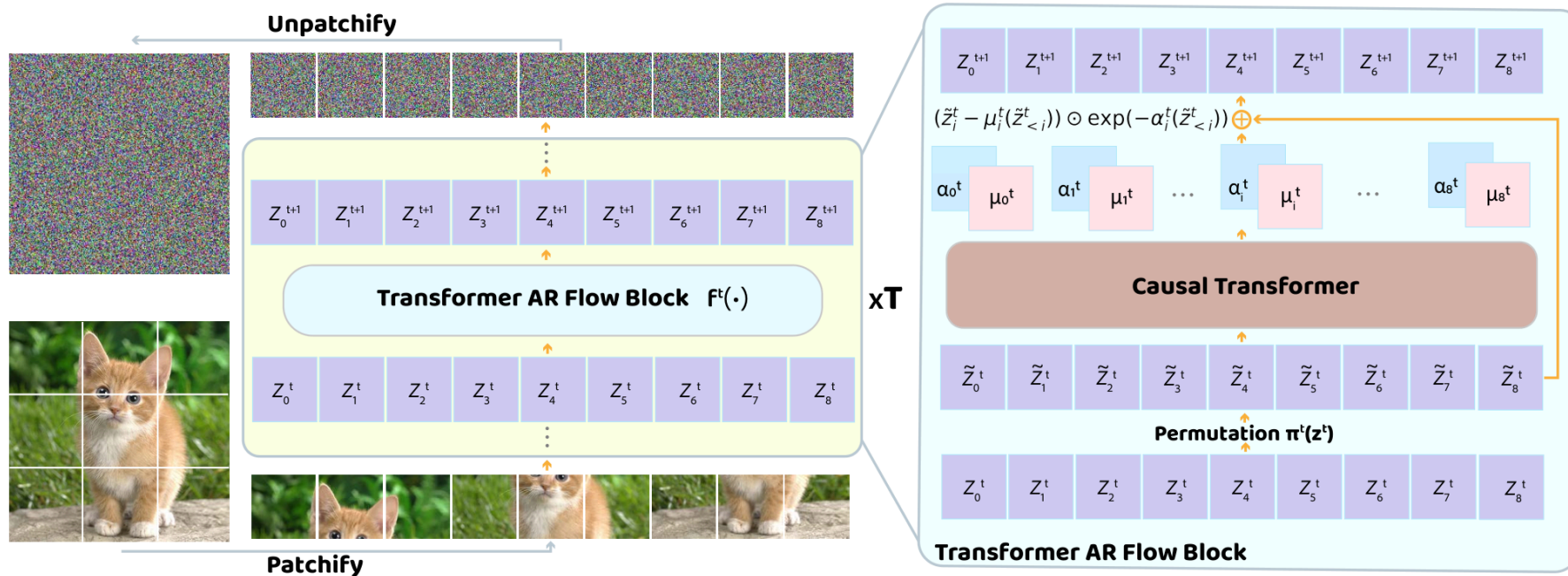
TarFlow: Transformer Architecture

Use a **causal Transformer** to predict the shifts and scales of each AR flow block.



TarFlow: Transformer Architecture

Use a **causal Transformer** to predict the shifts and scales of each AR flow block.



- Causal attention uses only earlier patches. The Transformer itself need not be invertible.
- Stack affine AR blocks and reverse patch order between blocks.

TarFlow: Flows Can Generate Detailed Images



AFHQ 256×256 (left); ImageNet 128×128 and 64×64 (right).

TarFlow: Flows Can Generate Detailed Images



AFHQ 256×256 (left); ImageNet 128×128 and 64×64 (right).

- The **autoregressive flow construction** scales to detailed images using networks over patches.
- The flow keeps a tractable likelihood and sequential sampling across patches.

Normalizing Flows: What Comes Next?

We can now turn simple noise into complex data, with both **sampling** and **exact density evaluation**.

What We Learned

- Change of variables gives an exact likelihood and an MLE training objective.
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Later: Continuous Flows (L9–L11)

Continuous-time normalizing flows (CNF) replace layers by an ODE, relaxing the layer constraints. Flow Matching learns the velocity field by regression; sampling uses a numerical solver.

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Bayesian Framework

Bayes' Theorem

$$p(\boldsymbol{\theta}|\mathbf{x}) = \frac{p(\mathbf{x}|\boldsymbol{\theta})p(\boldsymbol{\theta})}{p(\mathbf{x})} = \frac{p(\mathbf{x}|\boldsymbol{\theta})p(\boldsymbol{\theta})}{\int p(\mathbf{x}|\boldsymbol{\theta})p(\boldsymbol{\theta})d\boldsymbol{\theta}}$$

- $\mathbf{x}$: observed variables;
- $\boldsymbol{\theta}$: unknown latent variables/parameters;
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- $p(\mathbf{x}) = \int p(\mathbf{x}|\boldsymbol{\theta})p(\boldsymbol{\theta})d\boldsymbol{\theta}$: evidence;
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The prior $p(\boldsymbol{\theta})$ expresses our belief before observing data; the posterior $p(\boldsymbol{\theta}|\mathbf{x})$ updates it after observing $\mathbf{x}$.

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Computational Challenge

The evidence requires integration over the unobserved variables. If it is intractable, the posterior cannot be evaluated exactly.

Latent Variable Models (LVM)

Maximum Likelihood Estimation (MLE) Problem

For the observed dataset $\mathbf{X} = \{\mathbf{x}_i\}_{i=1}^n$,

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Latent Variable

Introduce a latent variable $\mathbf{z}$ for each observed sample $\mathbf{x}$:

$$p_{\boldsymbol{\theta}}(\mathbf{x}, \mathbf{z}) = p_{\boldsymbol{\theta}}(\mathbf{x}|\mathbf{z})p(\mathbf{z}); \quad \log p_{\boldsymbol{\theta}}(\mathbf{x}, \mathbf{z}) = \log p_{\boldsymbol{\theta}}(\mathbf{x}|\mathbf{z}) + \log p(\mathbf{z}).$$

Latent Variable Models (LVM)

Maximum Likelihood Estimation (MLE) Problem

For the observed dataset $\mathbf{X} = \{\mathbf{x}_i\}_{i=1}^n$,

$$\boldsymbol{\theta}^* = \arg \max_{\boldsymbol{\theta}} p_{\boldsymbol{\theta}}(\mathbf{X}) = \arg \max_{\boldsymbol{\theta}} \prod_{i=1}^n p_{\boldsymbol{\theta}}(\mathbf{x}_i) = \arg \max_{\boldsymbol{\theta}} \sum_{i=1}^n \log p_{\boldsymbol{\theta}}(\mathbf{x}_i).$$

The distribution $p_{\boldsymbol{\theta}}(\mathbf{x})$ should be highly complex (just like the true data distribution $p_{\text{data}}(\mathbf{x})$).

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- $p(\mathbf{z})$ is the **prior** and $p_{\boldsymbol{\theta}}(\mathbf{x}|\mathbf{z})$ is the **decoder** distribution.
- Both $p_{\boldsymbol{\theta}}(\mathbf{x}|\mathbf{z})$ and $p(\mathbf{z})$ are usually much simpler than $p_{\boldsymbol{\theta}}(\mathbf{x})$.

Latent Variable Models (LVM)

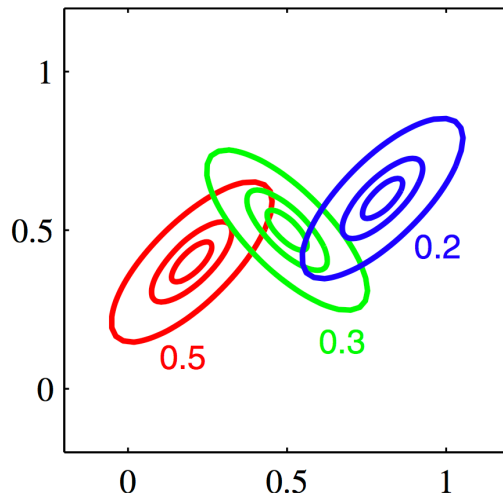
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Examples

Mixture of Gaussians



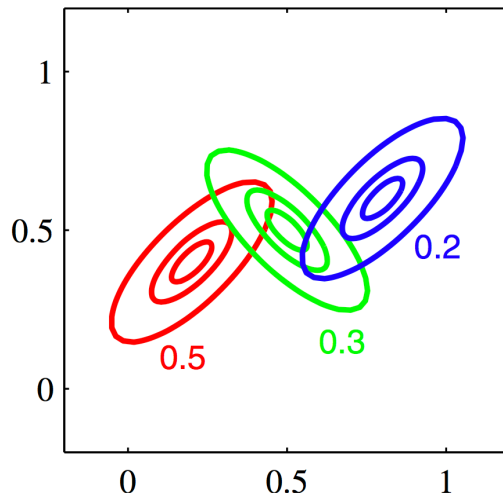
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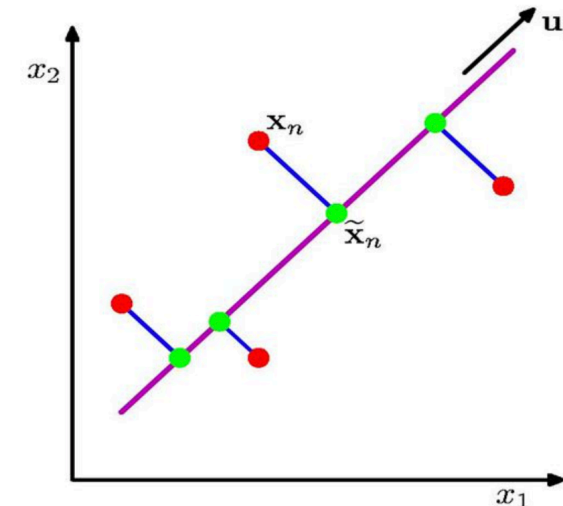
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PCA Model



- $p_{\theta}(\mathbf{x}|\mathbf{z}) = \mathcal{N}(\mathbf{W}\mathbf{z} + \boldsymbol{\mu}, \sigma^2\mathbf{I})$
- $p(\mathbf{z}) = \mathcal{N}(0, \mathbf{I})$

Summary

- Change of variables computes densities under invertible transformations.
- Normalizing flows use invertible maps with tractable Jacobians; CNF and Flow Matching extend this view of transporting distributions.
- Linear NFs use structured matrix factors to simplify determinant and inverse calculations.
- TarFlow uses Transformers to scale Gaussian AR flows with triangular Jacobians.
- RealNVP coupling is a special case of AR NF with fast density evaluation and sampling.
- LVMs combine a decoder $p_{\theta}(\mathbf{x}|\mathbf{z})$ and prior $p(\mathbf{z})$; marginalization makes $p_{\theta}(\mathbf{x})$ harder to evaluate.